Standard 12
PHYSICS
PART - I

Time: 3.00 Hours Marks: 70

Choose the best answer from the following: **15x1=15**

1) The dimension of $\frac{1}{\mu_0 \epsilon_0}$ is
a) $[LT^{-1}]$ b) $[L^2T^{-2}]$ c) $[L^{-1}T]$ d) $[L^{-2}T^2]$

2) A wire of length ℓ carrying a current I along the y direction is kept in a magnetic field given by $\vec{B} = \frac{\beta}{\sqrt{3}}(\hat{i}+\hat{j}+\hat{k})$ T. The magnitude of Lorentz force acting on the wire is
a) $\sqrt{\frac{2}{3}}\beta I\ell$ b) $\sqrt{\frac{1}{3}}\beta I\ell$ c) $\sqrt{2}\beta I\ell$ d) $\sqrt{\frac{1}{2}}\beta I\ell$

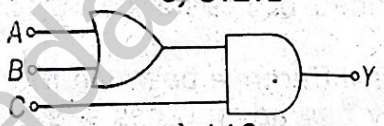
3) The temperature coefficient of resistance of a wire is 0.00125 per °C. At 20°C, its resistance is 1Ω. The resistance of the wire will be 2Ω at,
a) 800°C b) 700°C c) 850°C d) 820°C

4) When the current changes from +2A to −2A in 0.05 s an emf of 8V is induced in a coil. The coefficient of self induction of the coil is
a) 0.2 H b) 0.4 H c) 0.8 H d) 0.1 H

5) For light incident from air on a slab of refractive index 2, the maximum possible angle of refraction is
a) 30° b) 45° c) 60° d) 90°

6) The ratio of the wavelength radiation emitted for the transition from n = 2 to n = 1 in Li^{++}, He^+ and H is
a) 1:2:3 b) 1:4:9 c) 3:2:1 d) 4:9:36

7) The output of the following circuit is 1, when the input ABC is



a) 101 b) 100 c) 110 d) 010

8) First diffraction minimum due to a single slit of width 1.0×10^{-5} cm is at 30°. Then wavelength of light used is
a) 400 A° b) 500 A° c) 600 A° d) 700 A°

9) When a metallic surface is illuminated with radiation of wavelength λ, the stopping potential is V. If the same surface is illuminated with radiation of wavelength 2λ, the stopping potential is $\frac{V}{4}$. The threshold wavelength for the metallic surface is
a) 4λ b) 5λ c) $\frac{5}{2}\lambda$ d) 3λ

10) If an alternating voltage is represented as E = 141 sin 628 t the rms value of the voltage and frequency respectively
a) 141 V, 628 Hz b) 100 V, 50 Hz c) 100 V, 100 Hz d) 141 V, 100 Hz

11) The technology used for stopping the brain from processing pain is
a) precision medicine b) wireless brain sensor
a) virtual reality d) radiology

12) The radius of innermost orbit of hydrogen atom is 5.3×10^{-11} m. What is the radius of 3rd allowed orbit of hydrogen atom.
a) 4.77 A° b) 0.53 A° c) 1.06 A° d) 1.59 A°

13) The number of lines of force due to inc of positive charge in free space is
a) 10 b) 113 c) 11.29×10^{-6} d) 100

14) In interference pattern the width of the dark fringe is β_1, the width of the bright fringe is β_2, then
a) $2\beta_1 = \beta_2$ b) $2\beta_2 = \beta_1$ c) $\beta_1 = \beta_2$ d) $\beta_1 = \frac{1}{2}\beta_2$

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15) Two points A and B are maintained at a potential of 7V and −4V respectively. The work done in moving 50 electrons from A to B is
a) 8.80×10^{-17} J b) -8.80×10^{-17} J c) 4.40×10^{-17} J d) 5.80×10^{-17} J

PART - II

Answer any six of the following. Question No. 19 is compulsory: **6x2=12**

16) What is super conductors.
17) Mention the ways of producing induced emf.
18) Define capacitance. Give its unit.
19) Compute the speed of the electromagnetic wave in a medium if the amplitude of electric and magnetic fields are 3×10^{4} NC^{-1} and 2×10^{-4} T respectively.
20) State: Ampere's circuital law.
21) State de-Broglie hypothesis.
22) What are the shapes of wavefront for (a) Point source (b) Line source?
23) Mention the properties of neutrino.
24) In a transistor connected in the common base configuration, $\alpha = 0.95$, $I_E = 1$ m A, calculate the value of I_C and I_B.

PART - III

Answer any six of the following. Question No. 30 is compulsory: **6x3=18**

25) Obtain Gauss's law from Coulomb's law.
26) What is total internal reflection. Mention the conditions for total internal reflection to take place.
27) Give the properties of paramagnetic materials.
28) List out the laws of photoelectric effect.
29) Give two uses each of (a) IR radiation (b) Microwaves and (c) UV radiation
30) An ideal transformer has 460 and 40000 turns in the primary and secondary coils respectively. Find the voltage developed per turn of the secondary if the transformers is connected to a 230 V A.C. mains. The secondary is given to a load of resistance $10^{4}\Omega$. Calculate the power delivered to the load.
31) Mention the difference between interference and diffraction.
32) Mention any two advantages and disadvantages of Robotics.
33) When $^{235}_{92}U$ nucleus emits 2α, $3\beta^{-}$ and 2γ particles the atomic number and mass number of the resulting nucleus.

PART - IV

Answer all the following questions: **5x5=25**

34) Derive an expression for electrostatic potential due to an electric dipole.

(OR)

Discuss the spectral series of hydrogen atom.

35) Obtain the condition for bridge balance in wheatstones bridge.

(OR)

Explain about compound microscope and obtain the equation for the magnification.

36) Explain the construction and working of photoemissive cell. Mention any two uses of photocells.

(OR)

Obtain a relation for the magnetic field at a point due to an infinitely long straight conductor carrying current using Biot-Savart law.

37) Explain the types of absorption spectrum.

(OR)

Obtain lens makers formula and mention its significance.

38) State and prove De-morgans theorms.

(OR)

Explain the working of a single phase AC generator with necessary diagram.

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